

SANDIP DUTTA

+91-8961419611 | Bengaluru, India

sandip28dutta@gmail.com | linkedin.com/in/dutta-sd | github.com/Dutta-SD

EXPERIENCE

Software Development Engineer

Amazon

Jan 2024 – Present

Bengaluru, India

- Sole engineering owner of the delivery-time-prediction ML pipeline (design doc through launch) across **3 marketplaces**, integrating with **3 partner teams**; tightened customer delivery-promise accuracy by **~2 days**
- Diagnosed an upstream bottleneck on 20M+ records: re-tuned Spark joins (predicate pushdown, broadcast hints, disk caching) and rewrote the Pandas critical path in Polars, cutting a **30-min ETL to 5 min** and unblocking failing SageMaker jobs
- Migrated a customer-facing Java evaluation service off a legacy EC2 system onto ECS, gated by an AppConfig audit mode that shadow-executed the chain pre-cutover; introduced early-rejection filters at SQS ingress to skip fan-out for non-qualifying events, cutting SQS cost **~30%**
- Shipped a React-based seller-facing prompt to catch multi-item-order misses, saving **\$650K+ annually**
- Replaced a Glue+Athena+QuickSight reporting pipeline with an on-demand SQL self-serve flow, cutting recurring cost **~80%** and reclaiming **~5 SDE-weeks/yr** of operational toil
- Root-caused a recurring **5+ pages/week** on-call alarm class to test sellers bypassing production flows; introduced a flag check at the processor boundary, eliminating the ticket class entirely
- Built an MCP server over Amazon QuickSight letting seller associates self-serve flag-status queries across marketplaces, replacing a ticket-driven flow; serves **1000+ queries/day**, cutting ops load **~20%**

Research Software Engineer

IIT Kharagpur – VIP Lab

Jul 2023 – Jan 2024

Kharagpur, India

- Sole engineer for an end-to-end **CT segmentation pipeline** for radiotherapy planning; partnered with radiation oncologists from requirements through deployment, cutting per-case segmentation from **30 to 5 minutes** [paper]
- Drove rollout across **3 hospitals** including Tata Medical Center, Kolkata (**300+ scans/day** in production); Consulted on the on-prem to AWS migration
- Published 2 joint first-author papers: **FFLUNet** (**95% fewer parameters** for brain-tumor segmentation, *Computers in Biology and Medicine*) and **DualViT** (vision-transformer image classifier, *ICPR'24*)

Software Development Intern

AwesomeSuite Inc.

Jan 2023 – Jun 2023

Remote

- Cut API latency **40%** by resolving N+1 queries, optimizing SQL, and adding caching; built a document conversion service on AWS Lambda and an end-to-end digital signing workflow

Software Development Engineer Intern

Amazon

May 2022 – Jul 2022

Bengaluru, India

- Automated tax-document metadata extraction (eliminating **75%** of manual work, accelerating reporting **80%**) and cut data-pipeline runtime **40%** via optimized migration scripts

SKILLS

Languages & Databases

Java, Python, C++, SQL, DynamoDB, PostgreSQL, Redis

Frameworks & Backend

Spring Boot, FastAPI, REST APIs, Distributed Systems, gRPC, Spark

AI & Tooling

LLMs, MCP, LangChain, AWS, Docker, Git, CI/CD, JUnit, pytest

EDUCATION

Bachelor of Engineering (Hons.), Electronics and Telecommunication

Jadavpur University, Kolkata

Aug 2019 – Jun 2023

CGPA: 9.58 / 10.00